

Modelowanie i Identyfikacja

(MiI2024)

Materiał wykładowy: – Procesy Gaussowskie cz.2

Wydział Elektrotechniki i Automatyki

Katedra Inteligentnych Systemów Sterowania i Wspomagania Decyzji

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With $\mathbf{X} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_N \end{bmatrix}$, $f(\mathbf{X}) = \begin{bmatrix} f(x_1) \\ f(x_2) \\ \vdots \\ f(x_n) \end{bmatrix}$, $K(x_\star, x_\star) = \kappa(x_\star, x_\star)$,

$K(\mathbf{X}, x_\star) = \begin{bmatrix} \kappa(x_1, x_\star) \\ \vdots \\ \kappa(x_N, x_\star) \end{bmatrix} = K(x_\star, \mathbf{X})^\mathsf{T}$ and $K(\mathbf{X}, \mathbf{X}) = \begin{bmatrix} \kappa(x_1, x_1) & \dots & \kappa(x_1, x_N) \\ \vdots & & \vdots \\ \kappa(x_N, x_1) & \dots & \kappa(x_N, x_N) \end{bmatrix}$

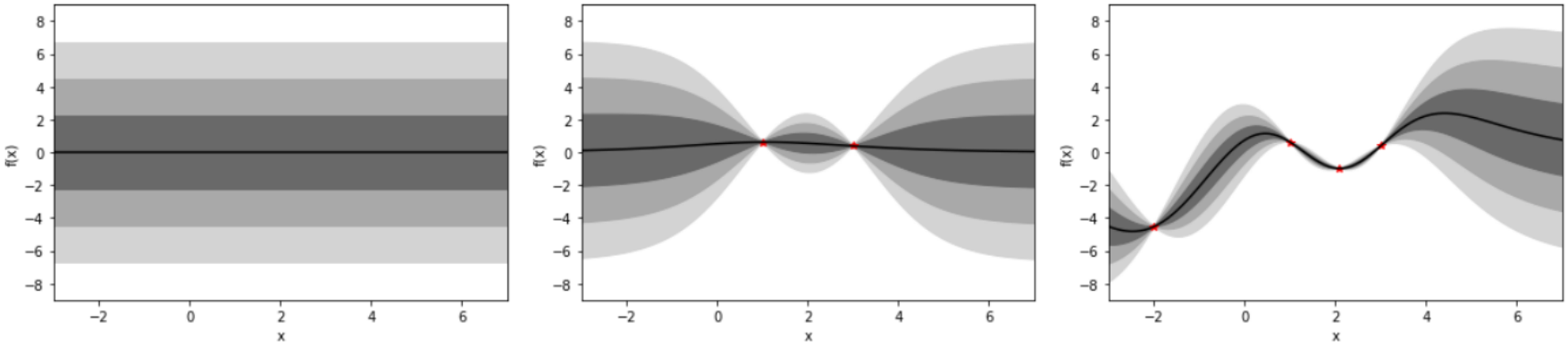
we have

$$\begin{bmatrix} f(\mathbf{X}) \\ f(x_\star) \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix} \mathbf{0} \\ 0 \end{bmatrix}, \begin{bmatrix} K(\mathbf{X}, \mathbf{X}) & K(x_\star, \mathbf{X}) \\ K(\mathbf{X}, x_\star) & K(x_\star, x_\star) \end{bmatrix} \right)$$

and most importantly

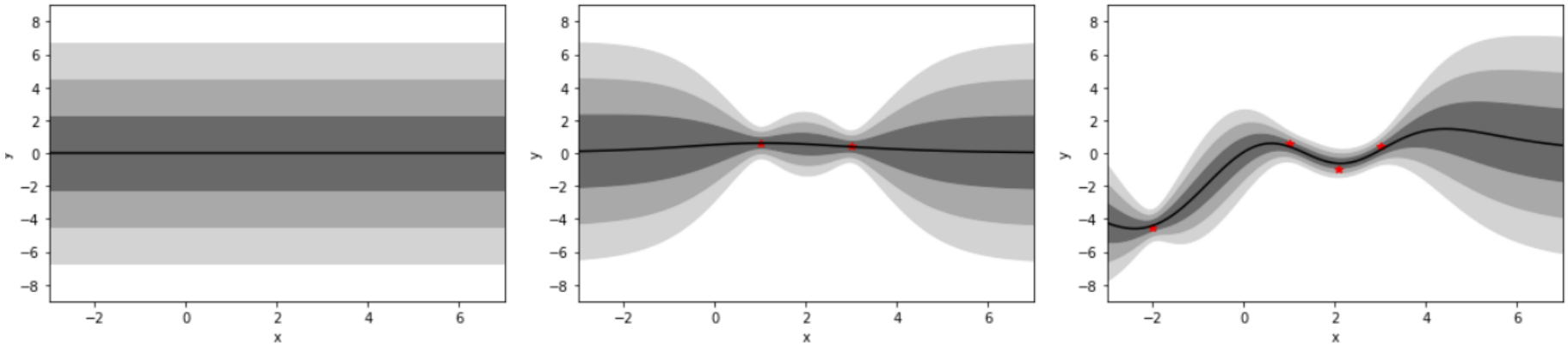
$$f(x_\star) \,|\, f(\mathbf{X}) \sim \mathcal{N}\left(\mathbf{K}(x_\star, \mathbf{X}) K(\mathbf{X}, \mathbf{X})^{-1} f(\mathbf{X}), K(x_\star, x_\star) - K(x_\star, \mathbf{X}) K(\mathbf{X}, \mathbf{X})^{-1} K(\mathbf{X}, x_\star) \right)$$

$$f(x_\star) \mid f(\mathbf{X}) \sim \mathcal{N}\left(\mathbf{K}(x_\star, \mathbf{X})K(\mathbf{X}, \mathbf{X})^{-1} f(\mathbf{X}), K(x_\star, x_\star) - K(\mathbf{X}, x_\star)K(\mathbf{X}, \mathbf{X})^{-1} K(\mathbf{X}, x_\star)\right)$$



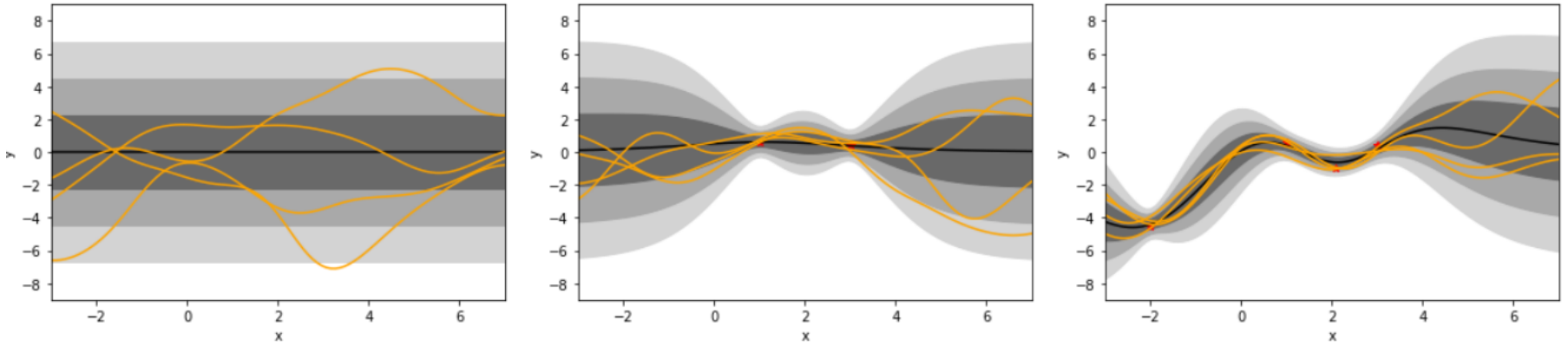
But what if we don't observe $f(x)$ exactly, but observe $f(x) + \varepsilon$, with $\varepsilon \sim \mathcal{N}(0, \sigma_n^2)$?

$$f(x_\star) \,|\, \mathbf{y} \sim \mathcal{N}\left(\mathbf{K}(x_\star, \mathbf{X})(K(\mathbf{X}, \mathbf{X}) + \sigma_n^2 \mathbf{I})^{-1}\mathbf{y}, K(x_\star, x_\star) - K(x_\star, \mathbf{X})(K(\mathbf{X}, \mathbf{X}) + \sigma_n^2 \mathbf{I})^{-1}K(\mathbf{X}, x_\star)\right)$$



We can also draw samples from $p(f(x_\star) \mid \mathbf{y})$

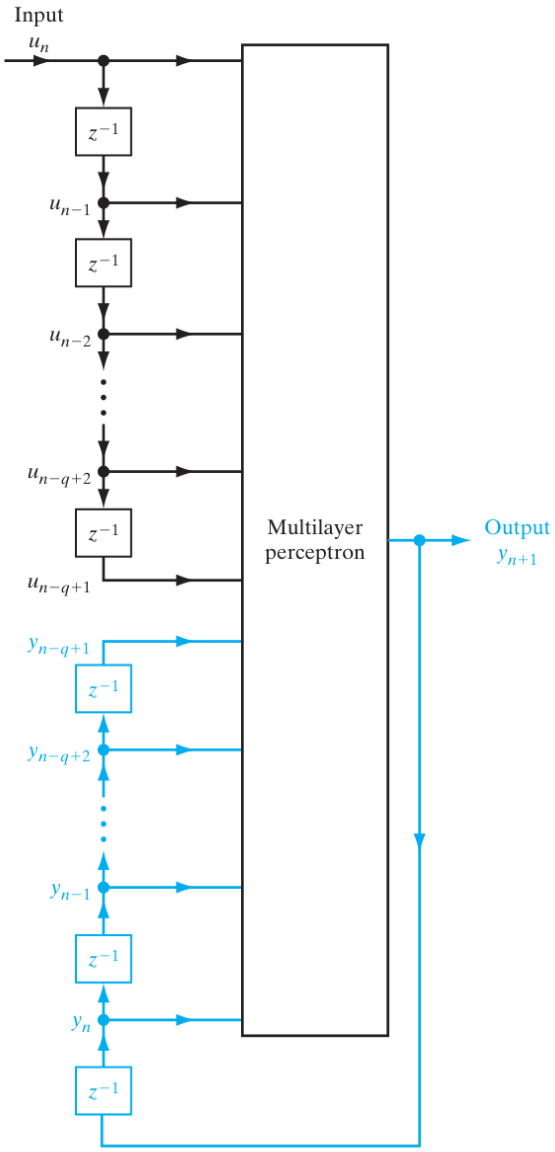
$$f(x_\star) \mid \mathbf{y} \sim \mathcal{N}\left(\mathbf{K}(x_\star, \mathbf{X})(K(\mathbf{X}, \mathbf{X}) + \sigma_n^2 \mathbf{I})^{-1}\mathbf{y}, K(x_\star, x_\star) - K(x_\star, \mathbf{X})(K(\mathbf{X}, \mathbf{X}) + \sigma_n^2 \mathbf{I})^{-1} K(\mathbf{X}, x_\star)\right)$$

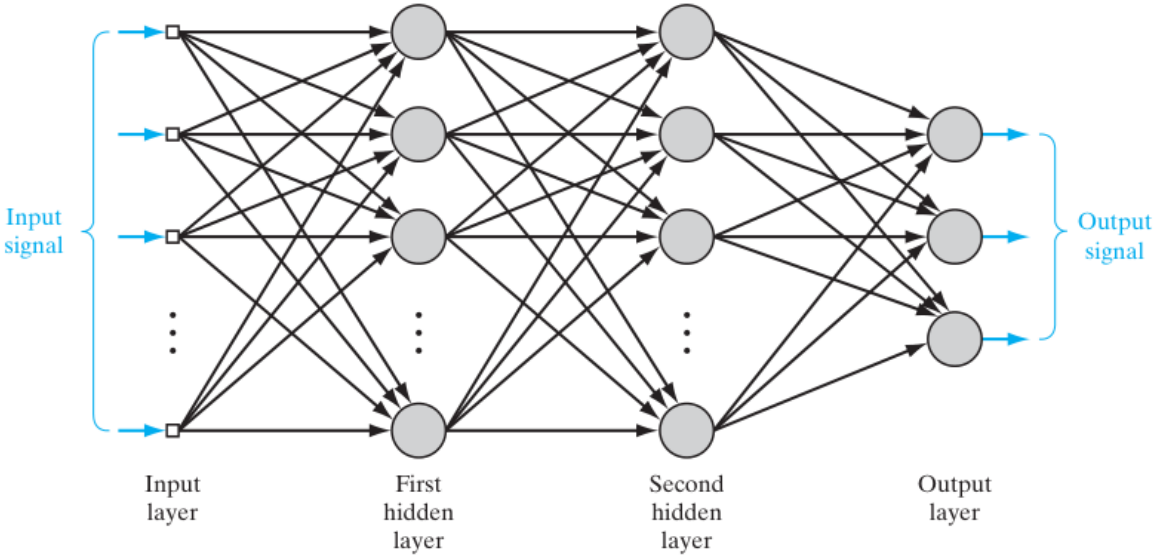


Do tej pory rozważaliśmy statyczne procesy Gaussowskie. W jaki sposób można wykorzystać procesy gaussowskie do modelowanie procesów dynamicznych?

NARX (nonlinear autoregressive model with exogenous input)

$$\hat{y}(k) = f(y(k-1), y(k-2), \ldots, y(k-n), u(k-1), u(k-2), \ldots, u(k-m)) + \nu,$$

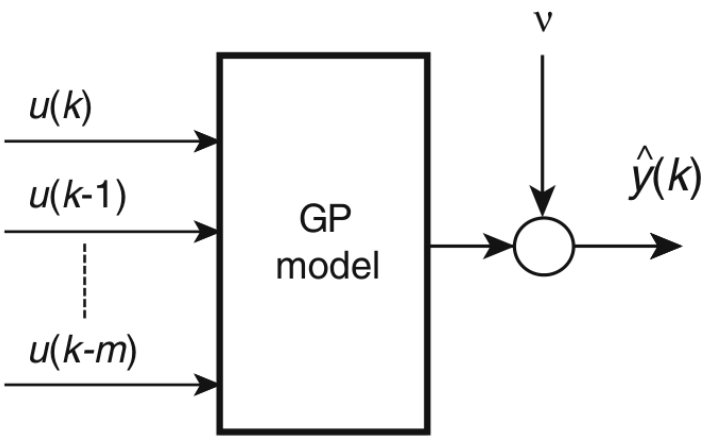




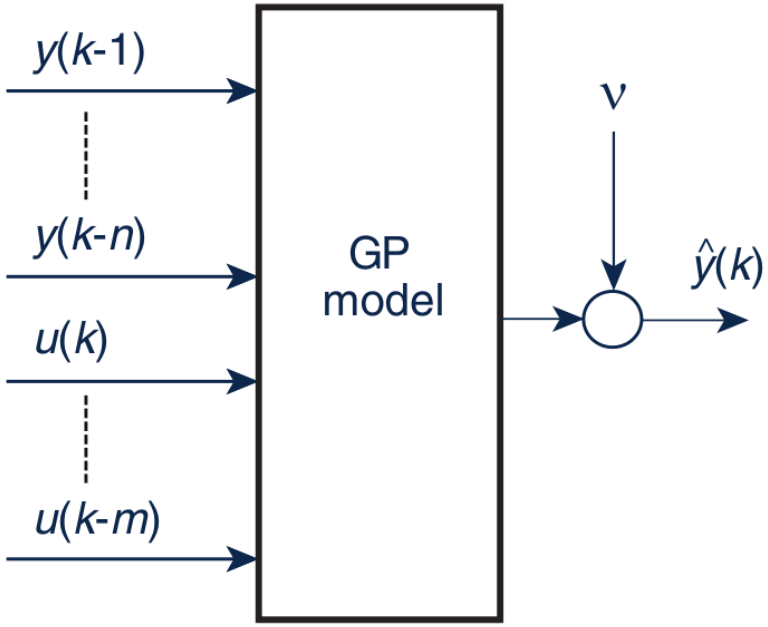
$$y_{n+1} = F(y_n, \ldots, y_{n-q+1}; u_n, \ldots, u_{n-q+1})$$

Analogia do sieci neuronowych typu NARX,

NFIR (nonlinear finite impulse response)



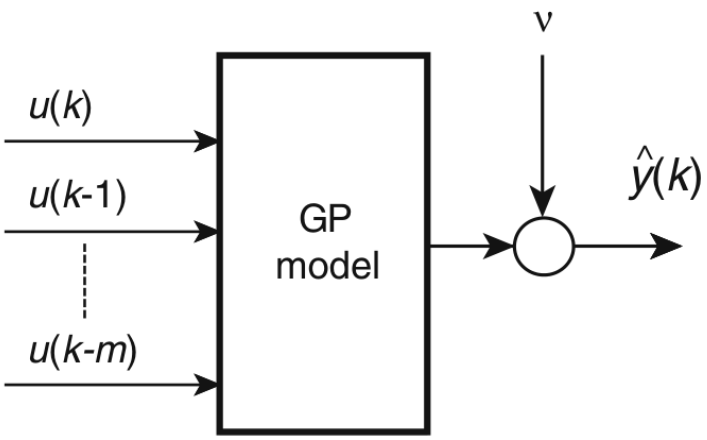
NARX (nonlinear autoregressive model with exogenous input)



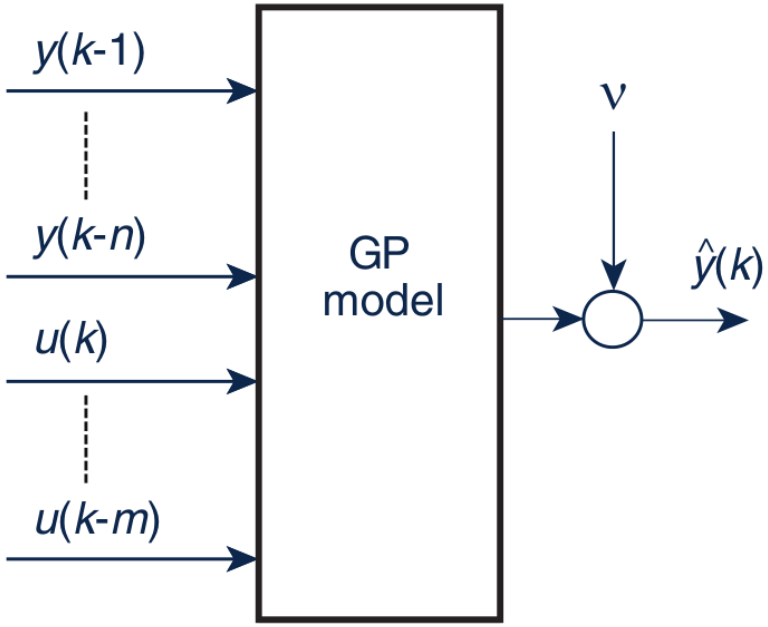
$$\hat{y}(k) = f(y(k-1), y(k-2), \ldots, y(k-n), u(k-1), u(k-2), \ldots, u(k-m)) + \nu,$$

Analogia do sieci neuronowych typu NARX,

NFIR (nonlinear finite impulse response)



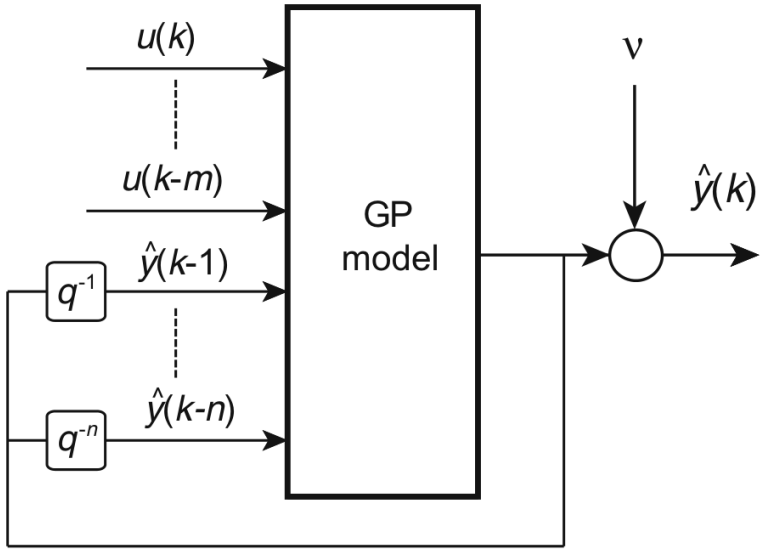
NARX (nonlinear autoregressive model with exogenous input)



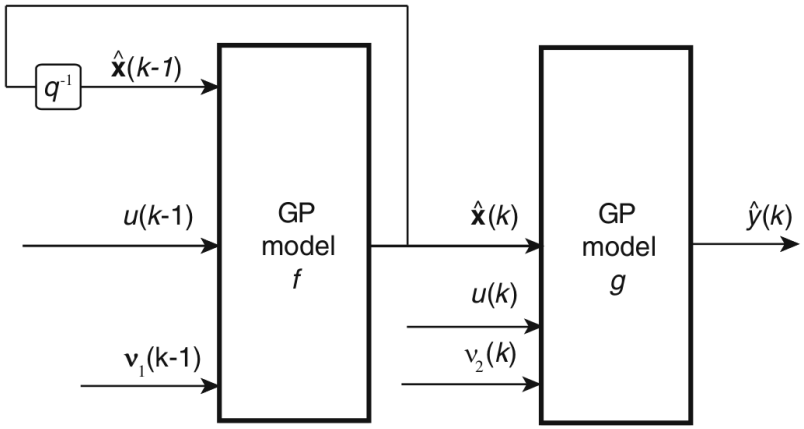
$$\hat{y}(k) = f(y(k-1), y(k-2), \ldots, y(k-n), u(k-1), u(k-2), \ldots, u(k-m)) + \nu,$$

Analogia do sieci neuronowych typu NARX,

NOE (nonlinear output error)



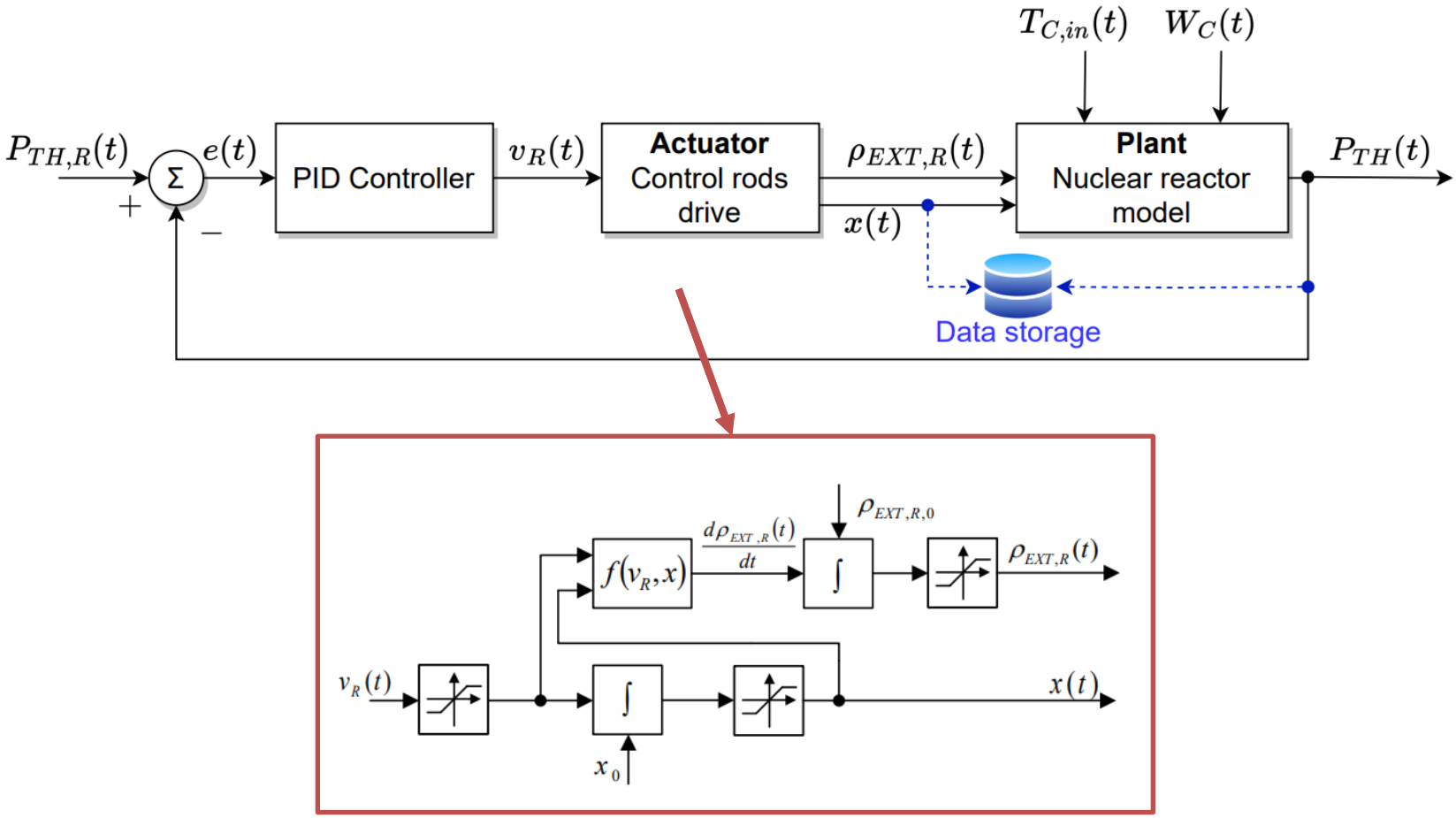
State-space model



$$\hat{y}(k) = f(\hat{y}(k-1), \hat{y}(k-2), \ldots, \hat{y}(k-n), u(k-1), u(k-2), \ldots, u(k-m)) + \nu.$$

$$\mathbf{x}(k+1) = f(\mathbf{x}(k), u(k)) + \boldsymbol{\nu}_1(k)$$
$$y(k) = g(\mathbf{x}(k), u(k)) + \nu_2(k)$$

Przykład zastosowania GP do identyfikacji i diagnostyki złożonego procesu dynamicznego



Przykład zastosowania GP do identyfikacji i diagnostyki złożonego procesu dynamicznego

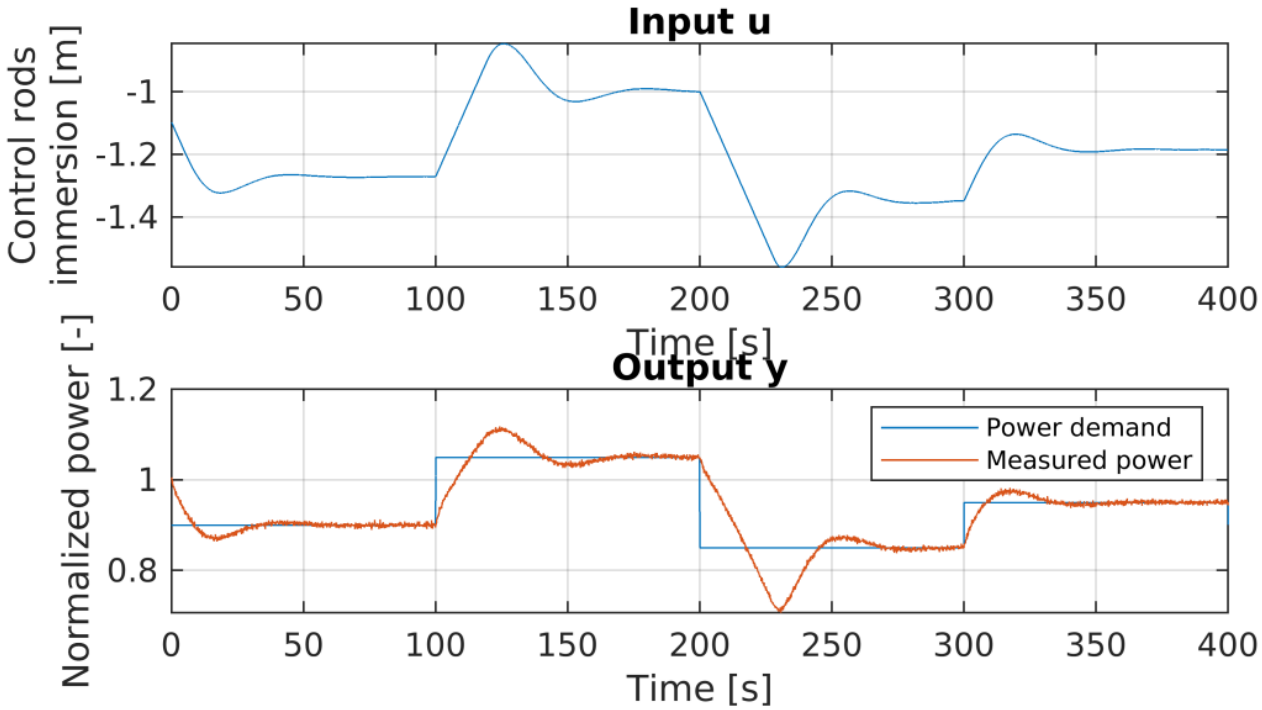


Fig. 4: Simulated data for GP model training.

Przykład zastosowania GP do identyfikacji i diagnostyki złożonego procesu dynamicznego

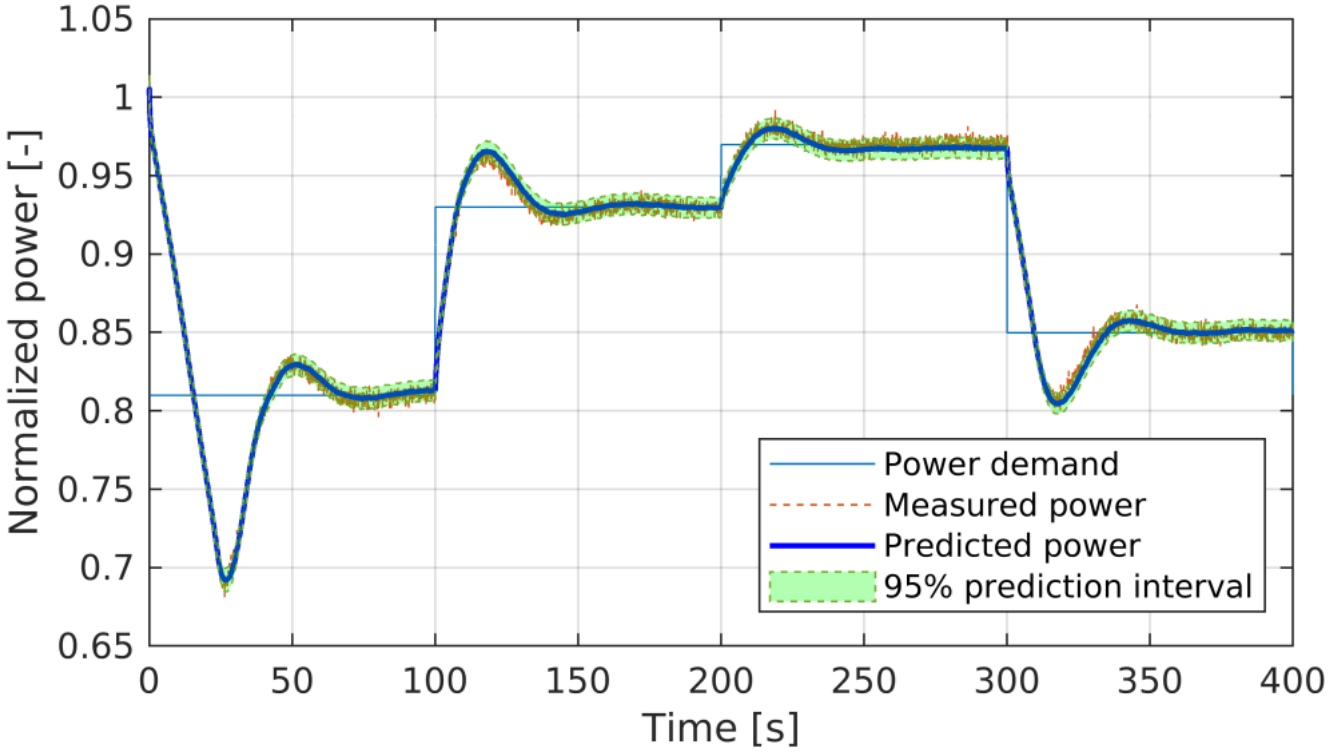


Fig. 5: Verification experiment of the trained GP model.

Przykład zastosowania GP do identyfikacji i diagnostyki złożonego procesu dynamicznego

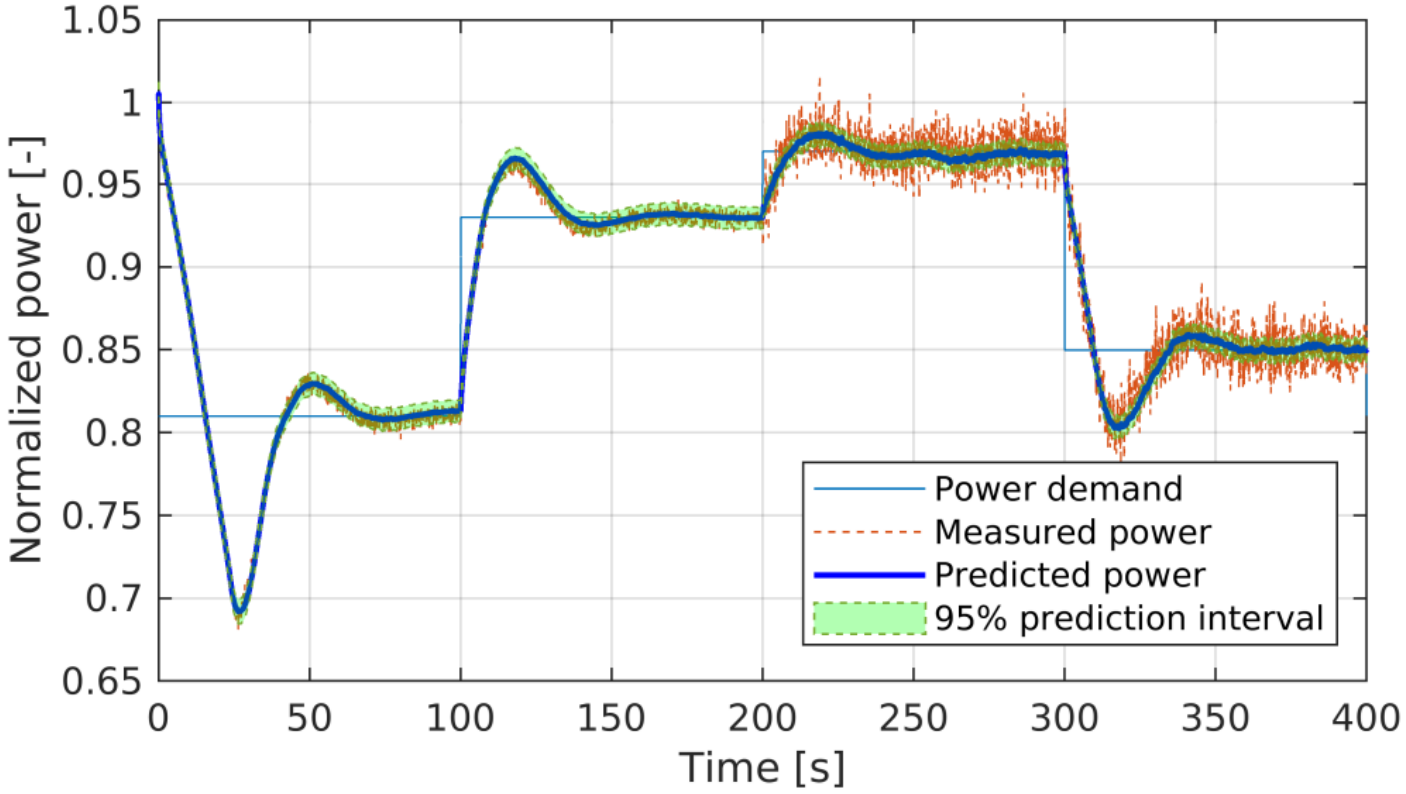


Fig. 6: Power monitoring system failure at $t = 200$s.

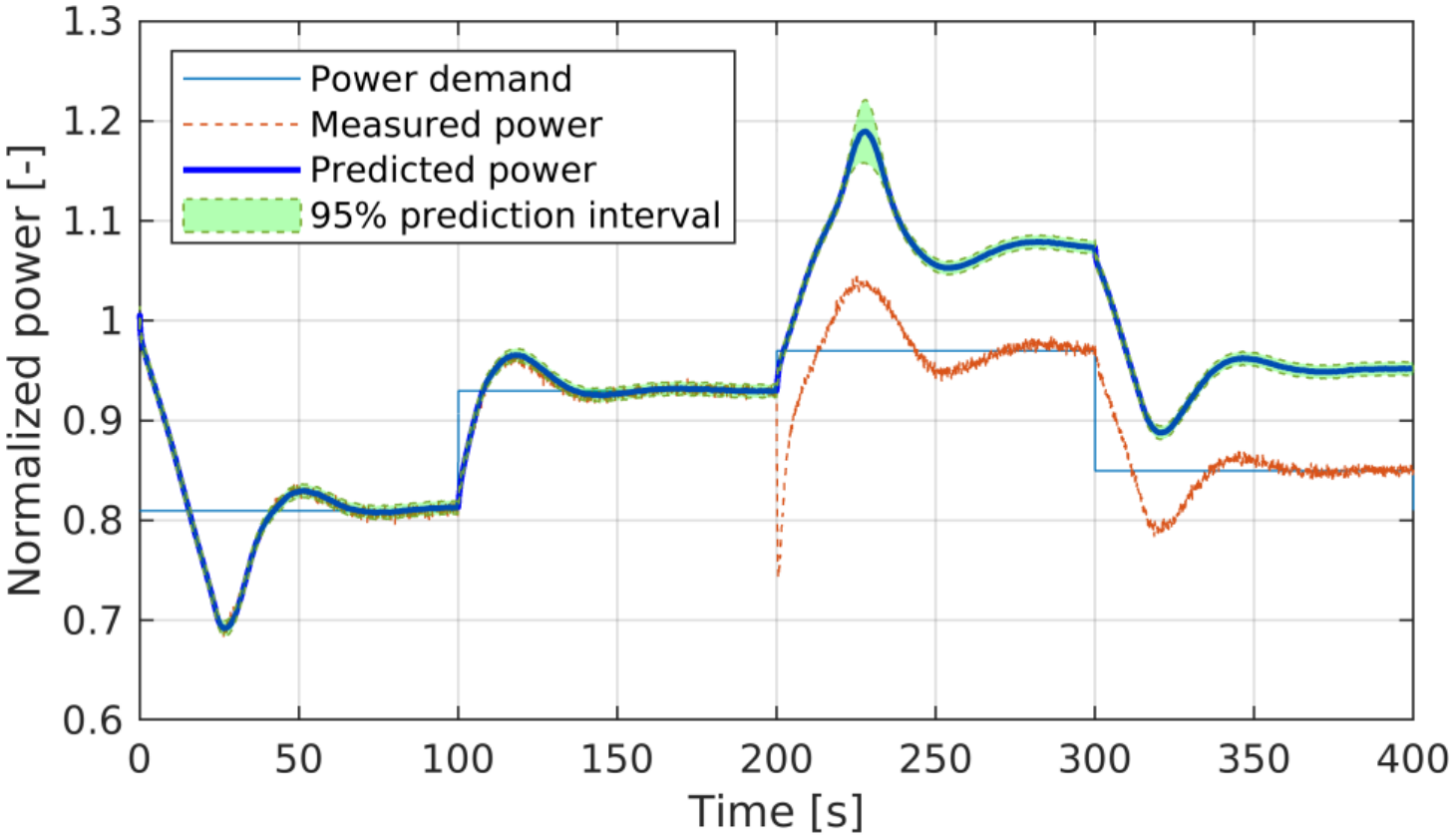


Fig. 7: Increase in coolant temperature at $t = 200$s.

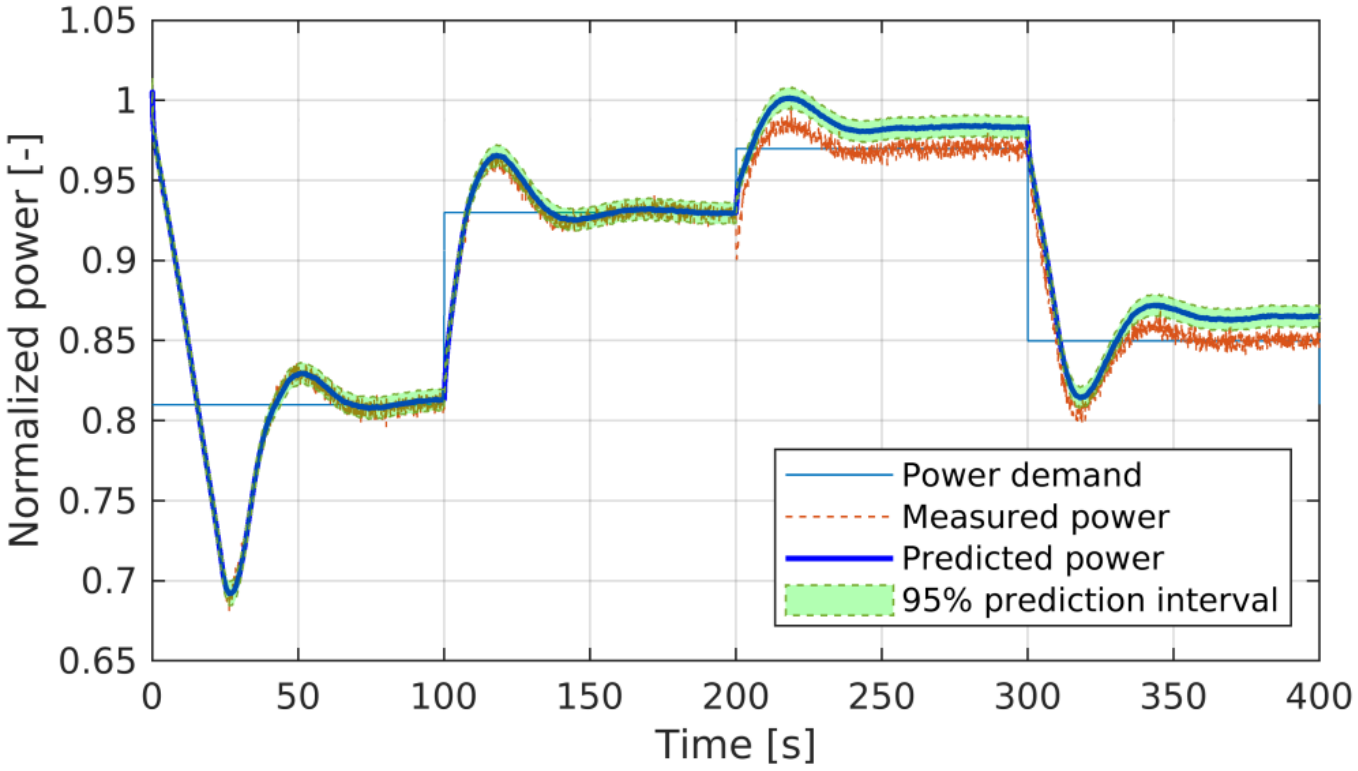


Fig. 8: Reduction of coolant flow at $t = 200$s.

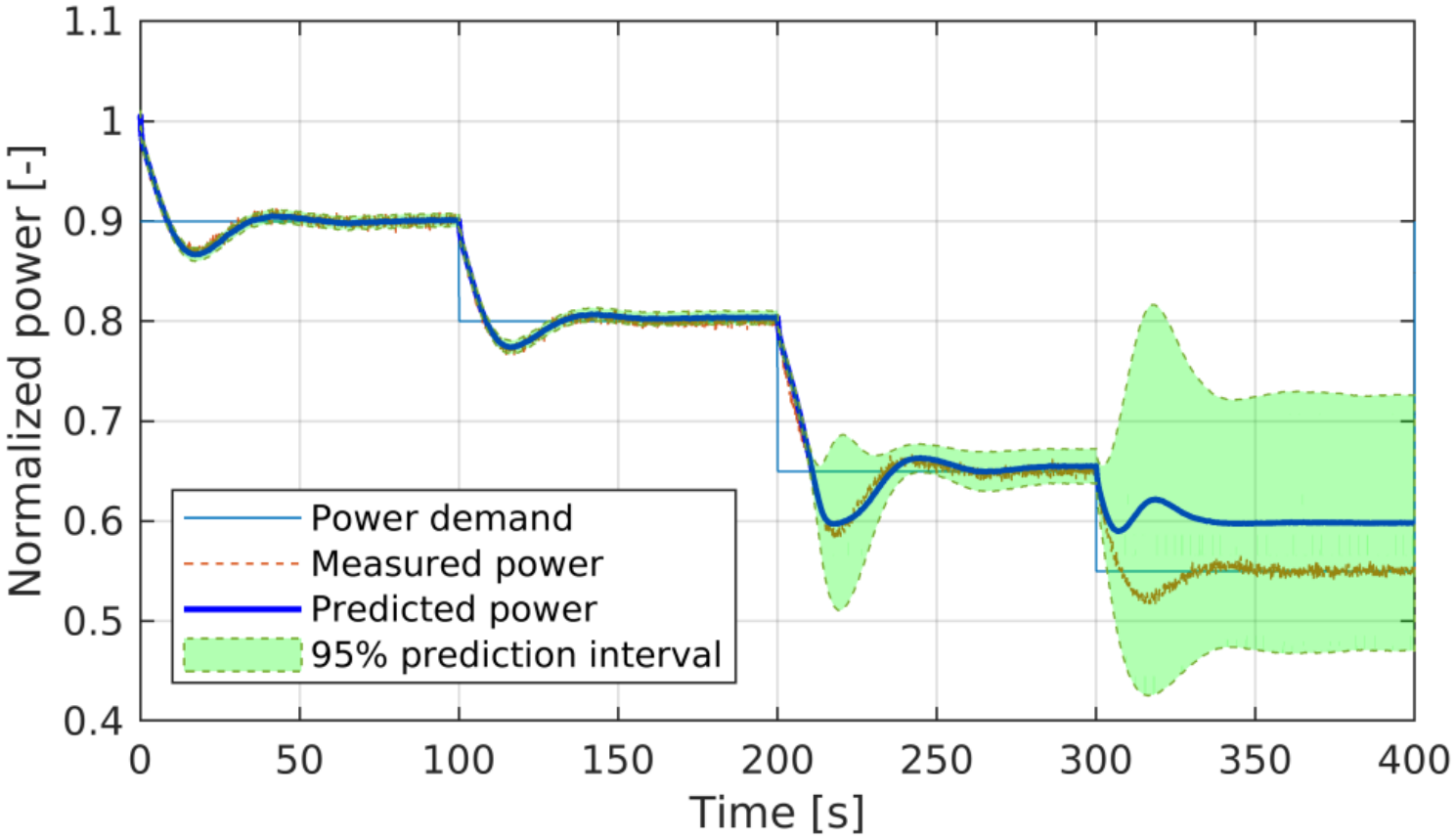


Fig. 9: Deviation from operating point at $t = 200$s.

Dziękuję
– koniec materiału prezentowanego podczas wykładu

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